

AI Architecture

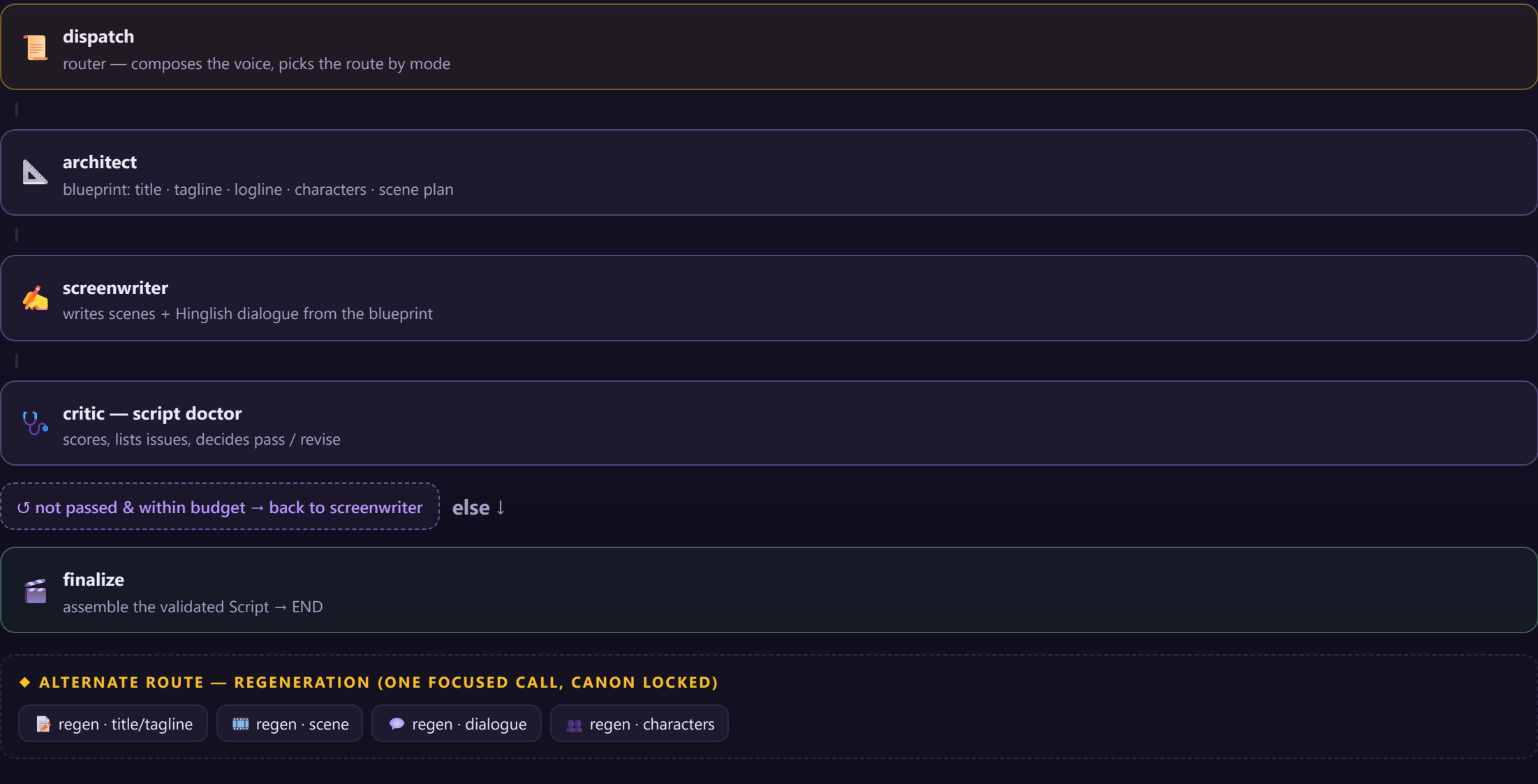
A LangGraph state machine drives three narrow agents through an always-on critic loop. Every model call funnels through one hardened Gemini client — rate-limited, retried, validated — and each completed node streams to the browser live over a WebSocket.



GRAPH/PIPELINE.PY — COMPILED STATEGRAPH

The orchestration graph

One shared **ScriptState** flows through every node; each node appends a trace entry that powers the live view.



LLM/CLIENT.PY — SINGLE CHOKE POINT

Every agent call, hardened

Agents stay simple; all resilience lives here.

- 1 Rate-limit gate**
Token bucket (RPM) + daily cap (RPD) before each call.
- 2 Thinking on**
Reasoning captured as thought summaries for observability.
- 3 Retry w/ backoff**
Up to 5× on 429 / 5xx, exponential + jitter.
- 4 Tolerant parse**
Strip <think> tags & fences, extract balanced JSON.
- 5 Validate + 1 repair**
Pydantic schema; one structured re-ask, then fail gracefully.



Structured handoffs

Pydantic models are the contract between every agent.

Live streaming

Each finished node emits a step event over the WebSocket.

3

narrow agents
+ 4 regen nodes

12 / 1400

RPM / RPD
self-throttle

Shared state

ScriptState carries blueprint, scenes, critique, trace.

Graceful degradation

Typed errors (rate-limit / agent) surface cleanly to the UI.

≤ 2

critic→revise loops
(loop guard)

1

Cloud Run instance
(accurate limiter)